Project 1 — E-Commerce Sales Intelligence Dashboard

DAY 2

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Objective

Run the ETL pipeline (etl_pipeline.py) against the 9 raw Olist CSV files, resolve all errors, and produce 6 clean output tables in data/processed/ — the star-schema foundation for the Power BI dashboard in Week 2.

Steps completed

#	Step	Detail	Status
1	**Place etl_pipeline.py in project root**	Downloaded fresh clean copy into C:\...\ecommerce-dashboard\etl_pipeline.py	**Done**
2	**First run — script exited silently, no output**	Ran python etl_pipeline.py — terminal returned to prompt immediately with no log lines printed.	**Issue found**
3	**Diagnosed: hidden characters from copy-paste**	Tested pandas independently: python -c "import pandas...". Got (99441,8) — confirmed environment is fine. Root cause: copy-paste from chat introduced invisible Unicode characters that caused silent failure.	**Issue found**
4	**Replaced with fresh clean file**	Generated new etl_pipeline.py written directly without copy-paste. Downloaded and replaced old file.	**Done**
5	**Second run — ETL succeeded**	python etl_pipeline.py ran fully. All 6 output CSVs saved to data/processed/. ETL complete message confirmed in terminal.	**Done**
6	**Verified output files in Explorer**	All 6 files visible in VS Code Explorer panel under data/processed/. Row counts confirmed.	**Done**

ETL output — row counts verified

All 6 tables written to `data/processed/` and confirmed in VS Code Explorer panel:

Table	File	Rows	Description
dim_date	`dim_date.csv`	**774**	One row per calendar day across the order date range
dim_customer	`dim_customer.csv`	**99,441**	All customers with surrogate key + location
dim_product	`dim_product.csv`	**32,951**	Products with English category + dimensions
dim_seller	`dim_seller.csv`	**3,095**	Sellers with location and surrogate key
dim_geography	`dim_geography.csv`	**19,015**	Zip codes with city, state, region, lat/lng
fact_orders	`fact_orders.csv`	**99,441**	Central fact table: revenue, freight, delivery, review score

Total rows processed across all tables: ~253,000 rows
ETL runtime: approximately 7 seconds | Log file: etl_log.txt created successfully

What the ETL pipeline does — function summary

`load_raw_data()`

Reads all 9 Olist CSVs from data/raw/ into pandas DataFrames. Logs row count per file.

`run_dq_checks()`

Scans every DataFrame for nulls and duplicates. Logs warnings. Applies documented fixes: fills missing delivered dates, fills missing product categories with 'unknown', fills review text nulls with empty string, deduplicates geolocation by zip code.

`build_dim_date()`

Generates a full date spine from min to max purchase date. Adds year, quarter, month, week, day_of_week, is_weekend columns. 774 rows covering the entire dataset date range.

`build_dim_customer()`

Renames raw columns to standard names. Assigns integer surrogate key (customer_key). Outputs customer_id, unique_id, zip_code, city, state.

`build_dim_product()`

Joins product table to category translation table. Fills untranslated categories. Assigns product_key surrogate. Outputs category_en, dimensions, weight.

`build_dim_seller()`

Renames and assigns seller_key surrogate. Outputs seller_id, zip, city, state.

`build_dim_geography()`

Deduplicates geolocation to one row per zip code. Maps state codes to regions (Southeast, South, Northeast, North, Central-West). Adds geo_key surrogate.

`build_fact_orders()`

Aggregates payments and reviews per order. Joins order items (first product/seller per order). Merges all dimensions via surrogate keys. Computes delivery_days, is_late flag, price_bucket. Outputs 17-column fact table ready for Power BI.

`save_tables()`

Writes all 6 tables to data/processed/ as UTF-8 CSVs. Logs row count per file.

Challenges faced and fixes applied

#	Challenge	Root cause	Fix
1	Script ran silently — no output, no log file, no error message	The file had hidden Unicode/zero-width characters introduced during copy-paste from the chat interface. Python parsed the file but the invisible characters prevented the logging handlers from initialising.	Deleted the file. Generated a fresh copy written directly to disk without copy-paste. Issue resolved on second run.

2	FutureWarning: inplace=True on DataFrame slice	Newer versions of pandas (2.x) deprecate inplace=True on chained operations like df[col].fillna(value, inplace=True). Triggers a FutureWarning.	Warning only — does not affect output. Will be refactored to df[col] = df[col].fillna(value) in the next iteration. Documented for Day 3.
3	Pylance showed 'pandas cannot be resolved' warnings in VS Code	VS Code selected the missing venv interpreter instead of Anaconda base, so Pylance could not find installed packages.	Selected correct interpreter via Ctrl+Shift+P > Python: Select Interpreter > Anaconda base path. Warnings are cosmetic only — ETL ran correctly regardless.
4	venv activation error on terminal open	VS Code tried to auto-activate a venv that does not exist in the project. This was a leftover VS Code workspace setting from the failed venv creation on Day 1.	Ignored — does not affect script execution. Anaconda base environment used directly. Will clean up workspace settings file if needed.

Key learnings from Day 2

1. Always test the environment independently (python -c "import pandas") before blaming the script.

2. Copy-paste from web/chat interfaces can introduce invisible Unicode characters — download files directly instead.

3. pandas 2.x deprecates inplace=True on chained operations — use df[col] = df[col].fillna() pattern going forward.

4. A silent ETL exit (no output, no error) almost always means the script crashed before the logging setup completed.

5. VS Code interpreter selection and actual script execution are independent — wrong interpreter shows warnings but script still runs via terminal python command.

6. Surrogate keys (integer range keys) in dimension tables are critical for Power BI relationships — natural string IDs cause slower joins.

Day 3 — next steps

Task	Detail
Create EDA notebook	notebooks/01_eda.ipynb — load all 6 processed CSVs using pandas
Revenue analysis	Monthly trend, top 10 categories by revenue, average order value
Delivery analysis	Avg delivery days by state, % late orders, on-time vs late distribution
Customer analysis	Repeat vs one-time buyers, review score distribution, payment type split
Annotate every chart	Add a markdown cell above each chart with a 1-sentence business insight with real numbers